ESP32 Camera Node Flow

cam_app_main.c - boot and common services

Power on
init_nvs()

device_identity_init()
Camera device ID

wifi_manager_init()
wait up to 15 s, retry

time_sync_configure_timezone()
/time sync until valid

cam_storage_init()
mount SD + create cam_undone/cam_done

cam_health_monitor_start()
independent heartbeat task

starts task

Boot camera backlog

cam_storage_count_done()
Any /sdcard/cam_done files?

yes /all uploaded

no

cam_upload_process_pending()
oldest RAW file first

Wi-Fi/server/ACK error

Upload failed
wait 10 s and retry

cam_record_init()
camera + capture/writer pipeline

Scheduled recording: 10 sec every 5 min

Cycle starts
CAM_RECORD_PERIOD_MS

cam_record_once()

cam_storage_begin_recording()
preallocate RAW file

capture_task - Core 1
DVP DMA -> PSRAM framebuffer
warm-up / frame validation

ready frame queue
zero-copy framebuffer pointers
no pixel copy into queue

writer_task - Core 0
copy PSRAM -> 32 KB internal DMA buffer
write() to SD in chunks

/sdcard/cam_undone/YYYYMMDD_HHMMSS.raw
recording in progress or failed

frame / FPS quality check
cam_storage_finish_recording()

pass

fail / dropped frames

rename undone -> done
GRAY8 640x480, target 5 FPS

Keep cam_undone
recording marked failed

skip upload

Independent camera health task

Collect Wi-Fi, heap, PSRAM, SD,
pending files, frame drops, upload stats

next period next period

POST /heartbeat
node_type=camera

/heartbeat

Flask server.py
/upload/video validates GRAY8 metadata
store RAW / convert to video / DB
return ACK JSON

Per-cycle upload

wifi_manager_wait_connected(15 s)

connected

offline

POST /upload/video
stream RAW in 4 KB chunks
headers: size, frames, width, height, FPS

ACK

HTTP 200 + ACK valid?
status + capture_id + bytes

more cam_done files

yes

no / timeout

cam_storage_delete_done()
remove local RAW

Keep cam_done file
retry on a later cycle

none left

Wait remaining period
then next cycle